SOME SCENARIOS

K8’s:

PROBES IN K8’s:

What is Probe:

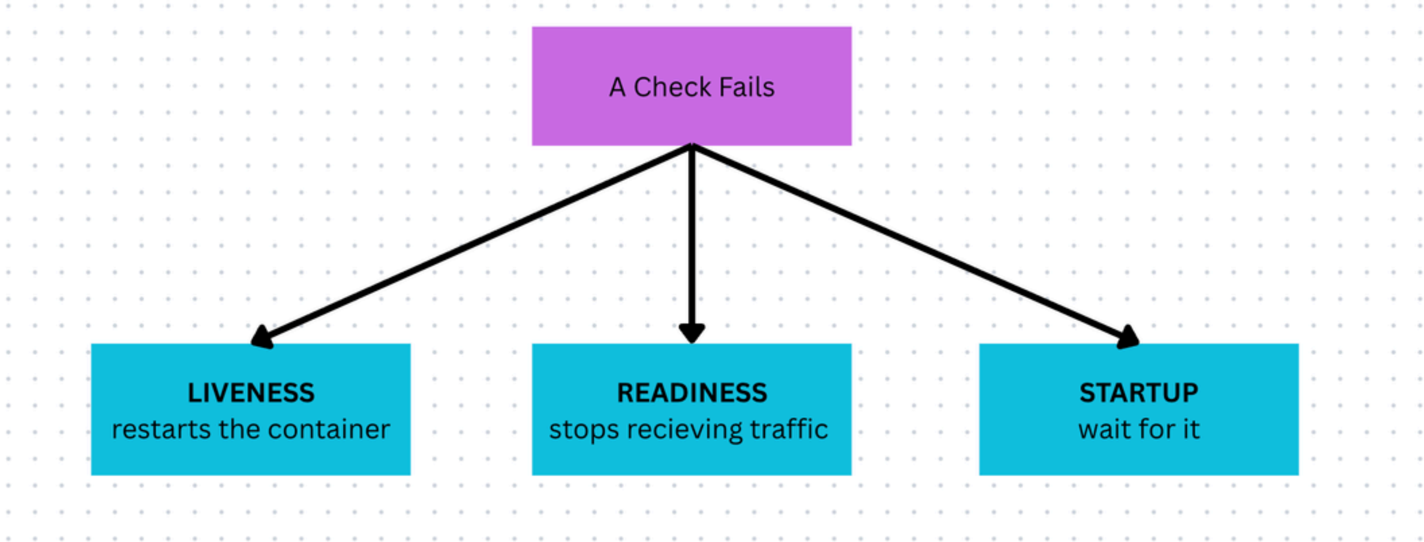
A probe is a small health check that k8s runs on your container again and again on a fixed schedule.

Remember: we never it by ourself. Kubelet on each node runs it for you automatically

Ex: kubelet is checking my container health status for every 5 seconds.

Question: But what if check fails?

Here what matters most is what k8s does when it fails?



Startup Probe (The wait action)

A pod can be created and the container can start, but sometimes the application inside takes a **long time to fully start**.

Example from real life:

You woke up from deep sleep.

That doesn’t mean you can immediately reply to messages or start office work.

Maybe first you:

- turn off the alarm
- wash your face
- drink coffee
- check notifications
- sit at your desk

Only after finishing this startup routine are you actually available.

Now imagine someone keeps asking every 5 seconds:

“Are you ready? Are you ready? Are you ready?”

You may look inactive even though you’re just starting.

That is exactly the problem **Startup Probe** solves.

In Kubernetes:

When the container starts:

- App may load huge datasets
- JVM may initialize
- Dependencies may start
- Database migrations may run
- Cache may build

This can take time.

If **Liveness Probe** starts checking immediately, it may think:

"Application is not responding → restart it"

Then:

- Container restarts
- Startup starts again
- Liveness kills it again

This becomes a restart loop.

Here comes Startup Probe

Startup Probe checks:

“Has the application completed startup?”

- If **startup probe fails**
  - Kubernetes waits

  - No restart by liveness yet
  - Readiness is also not considered
- If **startup probe succeeds**
  - Startup probe stops running
  - Then **Liveness Probe** starts
  - Then **Readiness Probe** starts

Readiness Probe (Traffic Control)

A pod can be alive (running) but not ready to accept traffic.

For example, when a person wakes up, they are awake but may not immediately start working. It can take some time to become active and productive.

Similarly, when a container starts:

- The application may still be loading data
- Cache may still be warming
- Dependencies may still be initializing
- Database connections may not be established

In this situation, Kubernetes uses a **Readiness Probe** to check whether the application is ready to receive requests.

- If the **readiness probe fails**:
  - Kubernetes keeps the pod **Running**
  - Removes the pod from the **Service endpoints**
  - Stops sending **new traffic** to that pod
  - Continues checking periodically
- Once the **readiness probe succeeds**:
  - Kubernetes marks the pod as **Ready**
  - Traffic starts flowing to that pod again

NOTE: **Readiness Probe never restarts the pod.** It only controls whether traffic is routed to it.

Liveness Probe (The Restart Action):

Sometimes the container is stuck (not crashed) due to

- Deadlock : 2 processes wait on each other
- Infinite loop : while loops
- Waiting on external service : Container → waiting for DB response but DB → unreachable

In this case, k8s cannot restart the container, Its just sits there, frozen

Here if we use **liveness** probe then kubelet kills the container and start the fresh one. Here only the container restart but pod stays

NOTE: use it when a restart can truly fix the problem.

Lets learn some core concetps:

InitialDelaySeconds: How long to wait for the very first check

PeriodSeconds: How often it checks. Default is 10 (it checks for every 10 sec)

timeoutSeconds: How long to wait for a reply before calling it failed

failure Threshold : How many failures before action happens?

Imagine your manager calls you.

- 1st call → no answer
- 2nd call → no answer
- 3rd call → manager decides:
- “Okay, something is wrong.”

That number (**3**) is the **failureThreshold**.

In Kubernetes:

failureThreshold: 3

means:

If the probe fails **3 times continuously**, Kubernetes takes action.

Action depends on probe type:

- **Startup Probe** → Container restarts
- **Liveness Probe** → Container restarts
- **Readiness Probe** → Stop sending traffic (pod keeps running)

Success Threshold : How many successes before Kubernetes trusts again?

Now imagine Wi-Fi dropped.

One signal bar appears.

You don’t immediately trust it.

You wait to see if connection is stable.

means:

Probe must pass **2 times continuously** before pod becomes Ready.

Flow:

Check 1 → Success
(Not ready yet)

Check 2 → Success
↓
Pod becomes Ready
Traffic allowed